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COMMON HALF YEARLY EXAMINATION - 2025

Standard XII

Reg.No. | | | | | | |

CHEMISTRY

Time : 3.00 hrs **Part – I** *Marks : 70*

I. Choose the correct answer: 15 x 1 = 15

1. Which of the following is used for concentrating ore in metallurgy?
   a) Leaching b) Roasting c) Froth floatation d) Both (a) and (c)
2. The basic structural unit of silicates is
   a) $(SiO_3)^{2-}$ b) $(SiO_4)^{2-}$ c) $(SiO)^{-}$ d) $(SiO_4)^{4-}$
3. On hydrolysis, PCl_3 gives
   a) H_3PO_3 b) PH_3 c) H_3PO_4 d) $POCl_3$
4. The magnetic moment of Mn^{3+} ion
   a) 4.80 BM b) 2.80 BM c) 8.95 BM d) 3.90 BM
5. A complex in which the oxidation number of the metal in zero is
   a) $K_4[Fe(CN)_6]$ b) $[Fe(CN)_3(NH_3)_3]$ c) $[Fe(CO)_5]$ d) both (b) and (c)
6. Assertion : Monoclinic sulphur is an example of monoclinic crystal system.
   Reason : For a monoclinic system $a \neq b \neq c$ and $\alpha = \gamma = 90°, \beta \neq 90°$
   a) Both assertion and reason are true and reason is the correct explanation of assertion.
   b) Both assertion and reason are true, but reason is not the correct explanation of assertion
   c) Assertion is true but reason is false
   d) Both assertion and reason are false
7. The rate constant of a reaction is $5.8 \times 10^{-2}\ s^{-1}$. The order of the reaction is
   a) First order b) Zero order c) Second order d) Third order
8. The pH of an aqueous solution is zero. The solution is
   a) slightly acidic b) strongly acidic c) neutral d) basic
9. The number of electrons that have a total charge of 9650 coulombs is
   a) 6.22×10^{23} b) 6.022×10^{24} c) 6.022×10^{22} d) 6.022×10^{-34}
10. Fog is colloidal solution of
    a) solid in gas b) gas in gas c) liquid in gas d) gas in liquid
11. Carbolic acid is
    a) phenol b) picric acid c) benzoic acid d) phenylacetic acid
12. In which of the following reactions new carbon-carbon bond is not formed ?
    a) Aldol condensation b) Friedel Craft reaction
    c) Kolbe's reaction d) Wolf Kishner reduction
13. Aniline + benzoylchloride $\xrightarrow{NaOH}$ $C_6H_5 - NH - COC_6H_5$.
    This reaction is known as
    a) Friedel–Crafts reaction b) HVZ reaction
    c) Schotten–Baumane reaction d) None of these
14. Which one of the following is not produced by body?
    a) DNA b) Enzymes c) Harmones d) Vitamins
15. Aspirin is a /an
    a) acetylsalicylic acid b) benzoyl salicylic acid
    c) chlorobenzoic acid d) anthranilic acid

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II. Answer any 6 questions. (Q.No.24 is compulsory) 6 x 2 = 12

16. What is Auto-reduction?
17. Describe briefly allotropism in p-block elements with specific reference to carbon.
18. What are interstitial compounds?
19. Write the cell reaction of Daniel cell.
20. What is Brownian movement ?
21. What is metamerism?
22. Give two difference between Hormones and Vitamins.
23. How is terylene prepared?
24. Establish a relationship between the solubility product and molar solubility for $Ag_2(CrO_4)$

Part - III

III. **Answer any 6 questions. (Q.No.33 is compulsory)** 6 x 3 = 18

25. How will you identify borate radical?
26. Give the uses of sulphuric acid.
27. What is crystal field stabilization energy (CFSE)?
28. Calculate the number of atoms in a fcc unit cell
29. Give the differences between order and molecularity of a reaction.
30. Write a note on catalytic poison.
31. Write Riemer-Tiemann reaction.
32. What are food preservatives?
33. Identify compounds A, B and C in the following sequence of a reaction.

$$C_6H_5NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[273\ K]{NaNO_2/HCl} B \xrightarrow[283\ K]{H_2O} C$$

Part - IV

IV. **Answer all the questions.** 5 x 5 = 25

34. a) Explain Zone refining process with an example. *(OR)*
    b) i) What type of hybridisation occur in (1) BrF_5 (2) BrF_3 (2)
    ii) (1) $XeOF_4 + SiO_2 \longrightarrow$ (2)
    (2) $Xe + F_2 \xrightarrow[400^\circ C]{Ni\ /\ 200\ atm}$ (1)
35. a) Compare Lanthanoids and Actinoids. *(OR)*
    b) In the complex $[Pt(NO_2)(H_2O)(NH_3)_2]Br$, Identify the following
    (i) Central metal atom / ion (ii) Ligand (s) and their types
    (iii) Coordination entity (iv) Oxidation number of the central metal ion
    (v) Coordination number
36. a) Explain Schottky and Frenkel defects. *(OR)*
    b) Derive integrated rate law for a first order reaction A $\longrightarrow$ product.
37. a) i) Discuss the Lowry–Bronsted concept of acids and bases (2)
    ii) Explain common ion effect with an example (3) *(OR)*
    b) How will you convert
    (i) Ethyl alcohol → Ethene (ii) Ethylene glycol → Ethanal (iii) Glycerol → Acrolein
38. a) Explain the reducing action of formic acid with example. *(OR)*
    b) How will you distinguish between primary, secondary and tertiary aliphatic amines.

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